active layers of the structure. The work of Coles *et al.* pushes the field further by offering the first evidence that such transfer can indeed take place via the hybrid modes of the system.

The ability to transfer energy over distances comparable to the wavelength of light has the potential to be of both fundamental and applied interest. At a fundamental level, our understanding of energy transfer underpins our ability to elucidate the processes responsible for photosynthesis in biological systems, as well as our ability to synthetically mimic their functionality. This analogy may be especially relevant given work suggesting that the energy transfer involved in photosynthesis may rely in part on a coherent coupling of molecules[1]. From an applied perspective, organic semiconductors continue to receive significant interest for application in optoelectronic devices, chief among these are light-emitting devices and photovoltaic devices. In both cases, optimum device performance is related to our ability to control the formation and transport of excitons in molecular systems. The work presented here may help researchers to think beyond existing paradigms by considering entirely new approaches to manipulating exciton behaviour. ❐

Russell J. Holmes is in the Department of Chemical Engineering and Materials Science, University of Minnesota, Minneapolis, Minneapolis 55455, USA.
e-mail: rholmes@umn.edu

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CRYSTAL NUCLEATION

Zeroing in on ice

Computer simulations show that cubic and hexagonal ices nucleate through the formation of a tetragonal metastable ice phase.

Ben Slater and David Quigley

By keeping it still, sufficiently pure water can be prevented from crystallizing when cooled well below its freezing temperature. In fact, supercooled liquids exist because the nucleation of a sufficiently large crystallite requires a large (and hence, extremely rare) fluctuation in energy. Indeed, for small crystallites, more energy is needed to form the crystal/liquid interface than that is gained from the transformation of the liquid into the (lower free energy) crystallite. Only when such unstable small crystallites grow large enough to be stable, spontaneous crystallization through homogenous nucleation occurs. Experiments[1] and computer simulations[1–3] have shown that the homogeneous crystallization of supercooled water leads to stackings of the stable cubic and hexagonal ices. Now, Hajime Tanaka and colleagues report in *Nature Materials* results from simulations that suggest that these stable ice nuclei form through the nucleation of a precursor metastable ice phase with tetragonal symmetry[4].

If a metastable crystal structure exists that is more amenable than stable crystal phases to forming interfaces with the liquid, then the formation of crystal nuclei of such metastable structure would be more likely to occur. In fact, the nucleation of metastable intermediates has been shown to occur in the freezing of a variety of model systems (an archetypal case is described in ref. 5) and is increasingly thought to be common in the precipitation of crystals from aqueous solutions (see, for example, ref. 6). Yet ice formation is believed to occur through the direct nucleation of the stable (and similar in free energy[7]) cubic and hexagonal phases despite that their homogeneous-nucleation line (the pressure-dependent temperature at which spontaneous nucleation can be readily observed in a pure sample) does not track their melting line, as the direct nucleation of these structures would suggest. Indeed, one would expect that the melting and homogeneous-nucleation curves of a nucleating crystal are parallel to each other. However, the identification of metastable structures for which the homogeneous-nucleation line follows the melting line and for which nuclei of a few hundred molecules persist only on submicrosecond timescales (that is, before they transform into a more stable phase) is experimentally challenging. Simulations of ice nucleation are however feasible for sufficiently simple water models of water.

In addition to 15 distinct stable crystalline phases of ice[8], ice structures that are metastable at conditions where hexagonal ice is thermodynamically favoured have been identified in computer simulations. However, the tetragonal ice structure proposed by Tanaka and co-authors (named ice 0), constructed by analogy to recently proposed semiconductor allotropes, does not seem to have been considered before. The authors find that ice 0 is a mechanically stable structure for two popular coarse-grained models of water, mW (ref. 9) and TIP4P (ref. 10). The structural analogy might raise an eyebrow, as mW, being an adaptation of a model for tetrahedrally coordinated semiconductor structures, models water as one effective site with energetically favoured tetrahedrally coordinated configurations. Still, TIP4P, a rigid four-site model, considers hydrogen bonding explicitly.

Quantum–mechanical methods can, however, offer a more stringent test of the energetic stability of a structure. We have thus calculated the relative stability (ignoring thermal contributions and with respect to hexagonal ice) of ice 0 as well as that of selected ice phases and alternative proposed structures[11–13] by using density functional theory, and compared the results with the values from the mW and TIP4P water models (Fig. 1). We have found that, among the proposed metastable structures, ice 0 is the closest in energy to hexagonal ice. Hence, our calculations support the conclusions of Tanaka and co-authors.

Moreover, the authors find that for the mW model the melting curve of ice 0 runs parallel to its homogeneous-nucleation line, and that during the nucleation process hexagonal ice emerges from the core of

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an ice 0 nucleus. In fact, they show that a two-parameter model based on classical nucleation theory, where hexagonal ice nucleates directly from the liquid, gives a relatively poor fit to free-energy simulation data of crystal nuclei as a function of size. However, an adapted three-parameter model that assumes that the stable phase emerges as a core of hexagonal ice within ice 0 provides a near perfect fit to the simulation data; still, the possibility that the improved fit to the simulation data arises purely by virtue of the extra fitting parameter remains to be checked. Direct calculation of these parameters, in particular the interfacial free energies between ice 0 and hexagonal ice, and ice 0 and the liquid, should provide additional support for the role of ice 0 in the ice-nucleation process as well as further understanding of the driving forces involved. Correctly capturing entropic contributions in such calculations and resolving the relative stabilities of ice phases below the 10 meV threshold are, however, significant challenges[14,15].

Although it is unlikely that ice 0 can be isolated as a pure phase, with quantitative predictions of its structural properties it may be possible to detect its signature by using high-frequency spectroscopy of samples in ice-crystallization experiments or of grain boundaries in polycrystalline hexagonal ice. The confirmation of the energetic stability of ice 0 represents an exciting new challenge to experiment and modelling, and potentially an important step in the understanding of the structure and thermodynamic behaviour of water. ❐

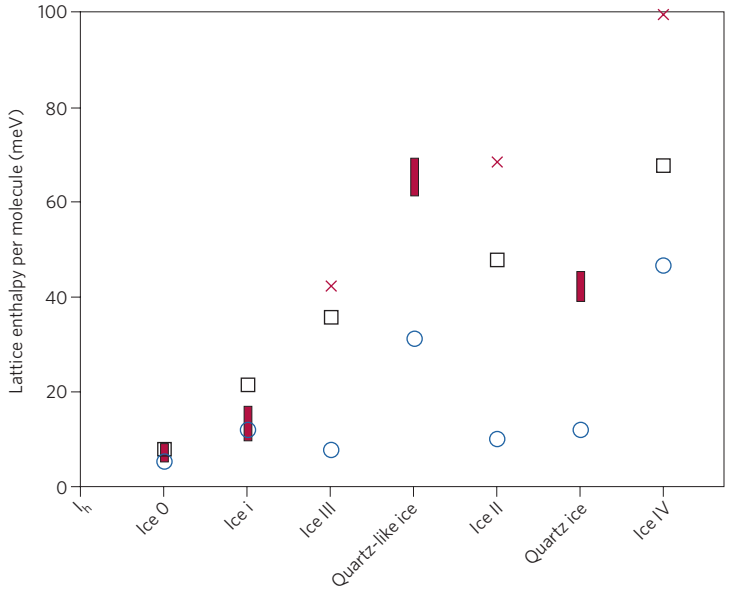


Figure 1 | Stability of several ice phases. Lattice enthalpies, with respect to that of ordinary hexagonal ice (I_h), of selected low-temperature ice phases (ices II, III and IV) and proposed ice structures (ice 0 (ref. 4), ice i (ref. 11), quartz-like ice[12] and quartz ice[13]) according to density functional theory (DFT) with the Perdew–Burke–Ernzerhof (PBE) functional (crosses and bars) and the classical mW (squares) and TIP4P (circles) water models. The bars denote the range of energies across all possible proton positions within the primitive (or unit) cell that satisfy the well-known 'ice-rules', where available. All PBE–DFT calculations were performed with version 7.0.2 of the CASTEP code using the plane-wave pseudopotential method (basis set was truncated at 480 eV, and the Brillouin zone was sampled at a spacing of 0.03 points per inverse ångström). TIP4P calculations were carried out in DL_POLY classic (12.5 Å real-space cutoff, with long-range electrostatic interactions handled via the Ewald method), and mW calculations were performed in LAMMPS. Calculations with functionals that explicitly include van der Waals interactions (PBEsol–D3 and PBE0–D3; results not shown) confirm that ice 0 is more stable than ice i and quartz ice.

Ben Slater is in the Department of Chemistry, University College London, 20 Gordon Street, London W C1H 0AJ, UK. David Quigley is in the Department of Physics and Centre for Scientific Computing, University of Warwick, Gibbet Hill Road, Coventry CV4 7AL, UK.
e-mail: b.slater@ucl.ac.uk; d.quigley@warwick.ac.uk

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SUPERCOOLED WATER

Continuous trends

X-ray scattering measurements of liquid water down to temperatures at which it spontaneously converts to ice show no signs of the much debated transition from high-density to low-density structural order.

Alan K. Soper

A bold proposal[1] for the explanation of some of water's unusual properties (for example, its density maximum at 4 °C) suggested water may have two critical points. A critical point — the most common being that at which coexisting vapour and liquid phases become indistinguishable — is the location in phase-space where several thermodynamic quantities, such as the compressibility and specific heat, diverge. At temperatures lower than the presumed second liquid–liquid critical point, water would exist as one of two liquids with different densities and measurably different structures. Unfortunately, the prediction — published more than two decades ago[1] and still debated today — is that the second

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